# 1 Second Fundamental Form

• Let $S : \mathbf{r} = \mathbf{r}(u,v)$ be a regular parametric surface, then the tangent space $T_pS$ at $p = \mathbf{r}(u_0,v_0) \in S$ is $T_pS = \mathrm{Span}_{\mathbb{R}}\{\partial_u\mathbf{r}(u_0,v_0), \partial_v\mathbf{r}(u_0,v_0)\}$ and the unit normal vector $\mathbf{n}$ at $p$ is given by

$$\mathbf{n} = \left.\frac{\partial_u\mathbf{r} \times \partial_v\mathbf{r}}{\|\partial_u\mathbf{r} \times \partial_v\mathbf{r}\|}\right|_{(u_0,v_0)}.$$

• Let $q = \mathbf{r}(u_0+\Delta u, v_0+\Delta v)$ be a nearby point of $p = \mathbf{r}(u_0,v_0) \in S$, then by the Taylor formula of multivariable vector valued functions, we get

$$\mathbf{r}(u_0+\Delta u, v_0+\Delta v) - \mathbf{r}(u_0,v_0)$$

$$= \begin{pmatrix} \partial_u\mathbf{r}(u_0,v_0) & \partial_v\mathbf{r}(u_0,v_0) \end{pmatrix} \begin{pmatrix} \Delta u \\ \Delta v \end{pmatrix} + \frac{1}{2} \begin{pmatrix} \Delta u & \Delta v \end{pmatrix} \begin{pmatrix} \partial_{uu}\mathbf{r}(u_0,v_0) & \partial_{uv}\mathbf{r}(u_0,v_0) \\ \partial_{vu}\mathbf{r}(u_0,v_0) & \partial_{vv}\mathbf{r}(u_0,v_0) \end{pmatrix} \begin{pmatrix} \Delta u \\ \Delta v \end{pmatrix} + \boldsymbol{o}\left((\Delta u)^2 + (\Delta v)^2\right),$$

where we adopt the notation that $\partial_{uv}\mathbf{r} := \frac{\partial^2\mathbf{r}}{\partial v \partial u}$ and we have $\partial_{uv}\mathbf{r} = \partial_{vu}\mathbf{r}$.

**Notation**: We denote

$$\begin{cases} L := \partial_{uu}\mathbf{r} \cdot \mathbf{n}, \\ M := \partial_{uv}\mathbf{r} \cdot \mathbf{n}, \\ N := \partial_{vv}\mathbf{r} \cdot \mathbf{n}. \end{cases}$$

Then the distance of the point $q = \mathbf{r}(u_0+\Delta u, v_0+\Delta v)$ to the tangent plane $T_pS$, which is denoted by $\delta(\Delta u, \Delta v)$, is given by

$$\delta(\Delta u, \Delta v) = \big(\mathbf{r}(u_0+\Delta u, v_0+\Delta v) - \mathbf{r}(u_0,v_0)\big) \cdot \mathbf{n}_p,$$

where $\mathbf{n}_p$ is the unit normal vector of $T_pS$, note that $\mathbf{n}_p \cdot \partial_u\mathbf{r} = \mathbf{n}_p \cdot \partial_v\mathbf{r} = 0$, so by some computation we get

$$\begin{aligned} \delta(\Delta u, \Delta v) &= \frac{1}{2} \begin{pmatrix} \Delta u & \Delta v \end{pmatrix} \left.\begin{pmatrix} L & M \\ M & N \end{pmatrix}\right|_{(u_0,v_0)} \begin{pmatrix} \Delta u \\ \Delta v \end{pmatrix} + \boldsymbol{o}\left((\Delta u)^2 + (\Delta v)^2\right) \\ &= \frac{1}{2}\left(L(\Delta u)^2 + 2M\Delta u\Delta v + N(\Delta v)^2\right) + \boldsymbol{o}\left((\Delta u)^2 + (\Delta v)^2\right). \end{aligned}$$

**Remark**: Since $\partial_u\mathbf{r} \cdot \mathbf{n} = \partial_v\mathbf{r} \cdot \mathbf{n} = 0$, thus $\partial_u\left(\partial_u\mathbf{r} \cdot \mathbf{n}\right) = \partial_{uu}\mathbf{r} \cdot \mathbf{n} + \partial_u\mathbf{r} \cdot \partial_u\mathbf{n} = 0$ and hence $L = -\partial_u\mathbf{r} \cdot \partial_u\mathbf{n}$. Similarly, we have $M = -\partial_u\mathbf{r} \cdot \partial_v\mathbf{n} = -\partial_v\mathbf{r} \cdot \partial_u\mathbf{n}$, $N = -\partial_v\mathbf{r} \cdot \partial_v\mathbf{n}$, which means

$$\begin{pmatrix} L & M \\ M & N \end{pmatrix} = -\begin{pmatrix} \partial_u\mathbf{r} \\ \partial_v\mathbf{r} \end{pmatrix} \begin{pmatrix} \partial_u\mathbf{n} & \partial_v\mathbf{n} \end{pmatrix}.$$

These discussion above yields the following definition of second fundamental form.

**Definition 1.(Second Fundamental Form)**
Let $S : \mathbf{r} = \mathbf{r}(u, v)$ be a regular parametric surface and let $\mathbf{n}$ be the unit vector field of $S$. Then the ***second fundamental form*** of $S$ is the the quadratic differential form defined on the tangent plane at each point by

$$\begin{aligned}\mathrm{II} &:= -\mathrm{d}\mathbf{r} \cdot \mathrm{d}\mathbf{n} = \mathrm{d}^2\mathbf{r} \cdot \mathbf{n} \\ &= L(\mathrm{d}u)^2 + 2M\mathrm{d}u\mathrm{d}v + N(\mathrm{d}v)^2 \\ &= \begin{pmatrix} \mathrm{d}u & \mathrm{d}v \end{pmatrix} \begin{pmatrix} L & M \\ M & N \end{pmatrix} \begin{pmatrix} \mathrm{d}u \\ \mathrm{d}v \end{pmatrix},\end{aligned}$$

where

$$\begin{cases} L := \partial_{uu}\mathbf{r} \cdot \mathbf{n} = -\partial_u\mathbf{r} \cdot \partial_u\mathbf{n}, \\ M := \partial_{uv}\mathbf{r} \cdot \mathbf{n} = -\partial_u\mathbf{r} \cdot \partial_v\mathbf{n} = -\partial_v\mathbf{r} \cdot \partial_u\mathbf{n}, \\ N := \partial_{vv} \cdot \mathbf{n} = -\partial_v\mathbf{r} \cdot \partial_v\mathbf{n}. \end{cases}$$

**Remark**: Let $u = u\big(\widetilde{u}, \widetilde{v}\big)$, $v = v\big(\widetilde{u}, \widetilde{v}\big)$ be an admissible change of parameters of the regular parametric surface $S : \mathbf{r} = \mathbf{r}(u, v)$, which preserves the orientation, i.e., $\frac{\partial(u,v)}{\partial(\widetilde{u},\widetilde{v})} > 0$, then we have

$$\partial_{\widetilde{u}}\mathbf{r} = \partial_u\mathbf{r} \cdot \frac{\partial u}{\partial \widetilde{u}} + \partial_v\mathbf{r} \cdot \frac{\partial v}{\partial \widetilde{u}}, \quad \partial_{\widetilde{v}}\mathbf{r} = \partial_u\mathbf{r} \cdot \frac{\partial u}{\partial \widetilde{v}} + \partial_v\mathbf{r} \cdot \frac{\partial v}{\partial \widetilde{v}},$$

hence

$$\partial_{\widetilde{u}}\mathbf{r} \times \partial_{\widetilde{v}}\mathbf{r} = \frac{\partial(u, v)}{\partial(\widetilde{u}, \widetilde{v})}\partial_u\mathbf{r} \times \partial_v\mathbf{r},$$

and it's clear that $\widetilde{\mathbf{n}} = \mathbf{n}$ and

$$\partial_{\widetilde{u}}\mathbf{n} = \partial_u\mathbf{n} \cdot \frac{\partial u}{\partial \widetilde{u}} + \partial_v\mathbf{n} \cdot \frac{\partial v}{\partial \widetilde{u}}, \quad \partial_{\widetilde{v}}\mathbf{n} = \partial_u\mathbf{n} \cdot \frac{\partial u}{\partial \widetilde{v}} + \partial_v\mathbf{n} \cdot \frac{\partial v}{\partial \widetilde{v}},$$

i.e.,

$$\begin{pmatrix} \partial_{\widetilde{u}}\mathbf{r} \\ \partial_{\widetilde{v}}\mathbf{r} \end{pmatrix} = \begin{pmatrix} \frac{\partial u}{\partial \widetilde{u}} & \frac{\partial v}{\partial \widetilde{u}} \\ \frac{\partial u}{\partial \widetilde{v}} & \frac{\partial v}{\partial \widetilde{v}} \end{pmatrix} \begin{pmatrix} \partial_u\mathbf{r} \\ \partial_v\mathbf{r} \end{pmatrix}, \quad \begin{pmatrix} \partial_{\widetilde{u}}\mathbf{n} \\ \partial_{\widetilde{v}}\mathbf{n} \end{pmatrix} = \begin{pmatrix} \frac{\partial u}{\partial \widetilde{u}} & \frac{\partial v}{\partial \widetilde{u}} \\ \frac{\partial u}{\partial \widetilde{v}} & \frac{\partial v}{\partial \widetilde{v}} \end{pmatrix} \begin{pmatrix} \partial_u\mathbf{n} \\ \partial_v\mathbf{n} \end{pmatrix},$$

so we adopt the notation that $J := \begin{pmatrix} \frac{\partial u}{\partial \widetilde{u}} & \frac{\partial v}{\partial \widetilde{u}} \\ \frac{\partial u}{\partial \widetilde{v}} & \frac{\partial v}{\partial \widetilde{v}} \end{pmatrix}$, i.e., $J$ is the transpose of the Jacobian matrix, thus we get

$$\begin{pmatrix} \widetilde{L} & \widetilde{M} \\ \widetilde{M} & \widetilde{N} \end{pmatrix} = -\begin{pmatrix} \partial_{\widetilde{u}}\mathbf{r} \\ \partial_{\widetilde{v}}\mathbf{r} \end{pmatrix} \begin{pmatrix} \partial_{\widetilde{u}}\mathbf{n} & \partial_{\widetilde{v}}\mathbf{n} \end{pmatrix} = -J \begin{pmatrix} \partial_u\mathbf{r} \\ \partial_v\mathbf{r} \end{pmatrix} \begin{pmatrix} \partial_u\mathbf{n} & \partial_v\mathbf{n} \end{pmatrix} J^{\mathrm{T}} = J \begin{pmatrix} L & M \\ M & N \end{pmatrix} J^{\mathrm{T}},$$

note that

$$\begin{pmatrix} \mathrm{d}u & \mathrm{d}v \end{pmatrix} = \begin{pmatrix} \mathrm{d}\widetilde{u} & \mathrm{d}\widetilde{v} \end{pmatrix} \begin{pmatrix} \frac{\partial u}{\partial \widetilde{u}} & \frac{\partial v}{\partial \widetilde{u}} \\ \frac{\partial u}{\partial \widetilde{v}} & \frac{\partial v}{\partial \widetilde{v}} \end{pmatrix} = \begin{pmatrix} \mathrm{d}\widetilde{u} & \mathrm{d}\widetilde{v} \end{pmatrix} J,$$

therefore

$$\begin{aligned}
\mathrm{II} &= L(\mathrm{d}u)^2 + 2M\mathrm{d}u\mathrm{d}v + N(\mathrm{d}v)^2 \\
&= \begin{pmatrix} \mathrm{d}u & \mathrm{d}v \end{pmatrix} \begin{pmatrix} L & M \\ M & N \end{pmatrix} \begin{pmatrix} \mathrm{d}u \\ \mathrm{d}v \end{pmatrix} \\
&= \begin{pmatrix} \mathrm{d}\widetilde{u} & \mathrm{d}\widetilde{v} \end{pmatrix} J \begin{pmatrix} L & M \\ M & N \end{pmatrix} J^{\mathrm{T}} \begin{pmatrix} \mathrm{d}\widetilde{u} \\ \mathrm{d}\widetilde{v} \end{pmatrix} \\
&= \begin{pmatrix} \mathrm{d}\widetilde{u} & \mathrm{d}\widetilde{v} \end{pmatrix} \begin{pmatrix} \widetilde{L} & \widetilde{M} \\ \widetilde{M} & \widetilde{N} \end{pmatrix} \begin{pmatrix} \mathrm{d}\widetilde{u} \\ \mathrm{d}\widetilde{v} \end{pmatrix} = \widetilde{\mathrm{II}},
\end{aligned}$$

which means that the second fundamental form is indepenent of the orientation preserving admissible change of parameters.

> **Example 1.** Find the first and second fundamental forms of a plane.

**Solution**: Let $S : \mathbf{r}(u,v) = (u,v,0), \quad (u,v) \in \mathbb{R}^2$ be the plane $Oxy$ in $\mathbb{R}^3$, then $\mathbf{n} = (0,0,1)$ and we get

$$\mathrm{d}\mathbf{r} = (\mathrm{d}u, \mathrm{d}v, 0), \quad \mathrm{d}\mathbf{n} = \mathbf{0},$$

therefore

$$\mathrm{I} = \mathrm{d}\mathbf{r} \cdot \mathrm{d}\mathbf{r} = (\mathrm{d}u)^2 + (\mathrm{d}v)^2, \quad \mathrm{II} = -\mathrm{d}\mathbf{r} \cdot \mathrm{d}\mathbf{n} = 0.$$

> **Example 2.** Find the first and second fundamental forms of a circular cylinder.

**Solution**: Let $S : \mathbf{r}(u,v) = (r\cos u, r\sin u, v)\,, \quad (u,v) \in [0,2\pi) \times \mathbb{R}$ be a circular cylinder, where $r > 0$ is a constant. Then we have

$$\partial_u \mathbf{r} = (-r\sin u, r\cos u, 0)\,, \quad \partial_v \mathbf{r} = (0,0,1),$$

therefore

$$\partial_u \mathbf{r} \times \partial_v \mathbf{r} = (r\cos u, r\sin u, 0), \quad \mathbf{n} = (\cos u, \sin u, 0),$$

$$\partial_{uu}\mathbf{r} = (-r\cos u, -r\sin u, 0), \quad \partial_{uv}\mathbf{r} = \partial_{vu}\mathbf{r} = \mathbf{0}, \quad \partial_{vv}\mathbf{r} = \mathbf{0},$$

hence we get

$$E = \|\partial_u \mathbf{r}\|^2 = r^2, \quad F = \partial_u \mathbf{r} \cdot \partial_v \mathbf{r} = 0, \quad G = \|\partial_v \mathbf{r}\|^2 = 1,$$

$$L = \partial_{uu}\mathbf{r} \cdot \mathbf{n} = -r, \quad M = \partial_{uv}\mathbf{r} \cdot \mathbf{n} = \mathbf{0}, \quad N = \partial_{vv}\mathbf{r} \cdot \mathbf{n} = 0,$$

so $\mathrm{I} = r(\mathrm{d}u)^2 + (\mathrm{d}v)^2, \quad \mathrm{II} = -r(\mathrm{d}u)^2$.

> **Example 3.** Let $S : \mathbf{r} = \mathbf{r}(u,v)$ be a regular parametric surface, then $S$ is a part of a plane $\Longleftrightarrow \mathrm{II} \equiv 0$.

**Proof**: ($\Rightarrow$) By Example 1;
($\Leftarrow$) It suffices to prove that $\mathbf{n}$ is a vector field of constant vectors. Since $\mathrm{II} \equiv 0$, so we have

$$L = -\partial_u \mathbf{r} \cdot \partial_u \mathbf{n} = 0, \quad M = -\partial_u \mathbf{r} \cdot \partial_v \mathbf{n} = -\partial_v \mathbf{r} \cdot \partial_u \mathbf{n} = 0, \quad N = -\partial_v \mathbf{r} \cdot \partial_v \mathbf{n} = 0.$$

By $\|\mathbf{n}\|^2 \equiv 1$, we have $\partial_u(\mathbf{n}\cdot\mathbf{n}) = \partial_v(\mathbf{n}\cdot\mathbf{n}) \equiv 0$, i.e., $\partial_v\mathbf{n}\cdot\mathbf{n} = \partial_v\mathbf{n}\cdot\mathbf{n} = 0$, note that $\{\mathbf{r}; \partial_u\mathbf{r}, \partial_v\mathbf{r}, \mathbf{n}\}$ is a frame on $S$, so we immediately get

$$\partial_u\mathbf{n} = \partial_v\mathbf{n} = \mathbf{0},$$

so

$$\mathrm{d}\mathbf{n} = \partial_u\mathbf{n}\cdot\mathrm{d}u + \partial_v\mathbf{n}\cdot\mathrm{d}v = 0,$$

which implies that $\mathbf{n}$ is a constant vector field. Note that $\mathrm{d}\mathbf{r}\cdot\mathbf{n} \equiv 0$, so

$$\mathrm{d}(\mathbf{r}\cdot\mathbf{n}) = \mathrm{d}\mathbf{r}\cdot\mathbf{n} + \mathbf{r}\cdot\mathrm{d}\mathbf{n} = 0,$$

i.e.,

$$\mathbf{r}\cdot\mathbf{n} \equiv \text{constant}.$$

Thus $\mathbf{r}(u,v)\cdot\mathbf{n} = \mathbf{r}(u_0,v_0)\cdot\mathbf{n}$ and hence $\big(\mathbf{r}(u,v) - \mathbf{r}(u_0,v_0)\big)\cdot\mathbf{n} = 0$, which is an equation for a plane. ■

> **Example 4.** Let $S : \mathbf{r} = \mathbf{r}(u,v)$ be a regular parametric surface, then $S$ is a part of a sphere $\Longleftrightarrow \mathrm{II} \triangleq c\cdot\mathrm{I}$ with $c \in \mathbb{R}^*$ is a constant.

**Proof**: ($\Rightarrow$) Let $S$ be a part of a sphere whoes center is $\mathbf{r}_0$ with radius $R$, i.e., $\big(\mathbf{r}(u,v) - \mathbf{r}_0\big)^2 = R^2$, then we get

$$\mathrm{d}\left(\big(\mathbf{r}(u,v) - \mathbf{r}_0\big)^2\right) = 0,$$

i.e., $\big(\mathbf{r}(u,v) - \mathbf{r}_0\big)\cdot\mathrm{d}\mathbf{r} = 0$, which means $\mathbf{r}(u,v) - \mathbf{r}_0$ is a normal vector of $S$ and hence $\mathbf{r}(u,v) - \mathbf{r}_0 = R\mathbf{n}$,

$$\mathrm{II} = -\mathrm{d}\mathbf{r}\cdot\mathrm{d}\mathbf{n} = -\mathrm{d}\mathbf{r}\cdot\mathrm{d}\left(\frac{1}{R}(\mathbf{r} - \mathbf{r}_0)\right) = -\frac{1}{R}\mathrm{d}\mathbf{r}\cdot\mathrm{d}\mathbf{r} = -\frac{1}{R}\cdot\mathrm{I};$$

($\Leftarrow$) Suppose there is a fucntion $f(u,v)$ such that

$$\mathrm{II} = f(u,v)\cdot\mathrm{I}$$

and $f(u,v) \neq 0$ for $\forall(u,v)$. So we have

$$\big(L - f(u,v)E\big)(\mathrm{d}u)^2 + 2\big(M - f(u,v)F\big)\mathrm{d}u\mathrm{d}v + \big(N - f(u,v)G\big)(\mathrm{d}v)^2 \equiv 0,$$

i.e.,

$$\begin{cases} \partial_u\mathbf{r}\cdot\big(\partial_u\mathbf{n} + f(u,v)\partial_u\mathbf{r}\big) = 0, \\ \partial_v\mathbf{r}\cdot\big(\partial_u\mathbf{n} + f(u,v)\partial_u\mathbf{r}\big) = \partial_u\mathbf{r}\cdot\big(\partial_v\mathbf{n} + f(u,v)\partial_v\mathbf{r}\big) = 0, \\ \partial_v\mathbf{r}\cdot\big(\partial_v\mathbf{n} + f(u,v)\partial_v\mathbf{r}\big) = 0. \end{cases}$$

Note that $\partial_u\mathbf{n}\cdot\mathbf{n} = \partial_v\mathbf{n}\cdot\mathbf{n} = 0$ since $\|\mathbf{n}\| = 1$ and $\partial_u\mathbf{r}\cdot\mathbf{n} = \partial_v\mathbf{r}\cdot\mathbf{n}$=0, so we also get

$$\begin{cases} \mathbf{n}\cdot\big(\partial_u\mathbf{n} + f(u,v)\partial_u\mathbf{r}\big) = 0, \\ \mathbf{n}\cdot\big(\partial_v\mathbf{n} + f(u,v)\partial_v\mathbf{r}\big) = 0. \end{cases}$$

Since $\{\mathbf{r}; \partial_u\mathbf{r}, \partial_v\mathbf{r}, \mathbf{n}\}$ is a frame on $S$, so we immediately get

$$\partial_u\mathbf{n} + f(u,v)\partial_u\mathbf{r} = \partial_v\mathbf{n} + f(u,v)\partial_v\mathbf{r} = \mathbf{0},$$

thus

$$\partial_v\big(\partial_u\mathbf{n}+f(u,v)\partial_u\mathbf{r}\big)=\partial_{uv}\mathbf{n}+\frac{\partial f(u,v)}{\partial v}\partial_u\mathbf{r}+f(u,v)\partial_{uv}\mathbf{r}=0,$$

$$\partial_u\big(\partial_v\mathbf{n}+f(u,v)\partial_v\mathbf{r}\big)=\partial_{vu}\mathbf{n}+\frac{\partial f(u,v)}{\partial u}\partial_v\mathbf{r}+f(u,v)\partial_{vu}\mathbf{r}=0,$$

so we see that

$$\frac{\partial f(u,v)}{\partial v}\cdot\partial_u\mathbf{r}=\frac{\partial f(u,v)}{\partial u}\cdot\partial_v\mathbf{r},$$

which implies that

$$\frac{\partial f}{\partial u}=\frac{\partial f}{\partial v}=0$$

because $\{\partial_u\mathbf{r},\partial_v\mathbf{r}\}$ is linearly independent. So $f(u,v)\triangleq c\in\mathbb{R}^*$ is a constant, i.e., $\mathrm{II}=c\cdot\mathrm{I}$. This means

$$\partial_u\mathbf{n}+c\cdot\partial_u\mathbf{r}=\partial_v\mathbf{n}+c\cdot\partial_v\mathbf{r}=\mathbf{0},$$

hence

$$\mathrm{d}(\mathbf{n}+c\cdot\mathbf{r})=\big(\partial_u\mathbf{n}+c\cdot\partial_u\mathbf{r}\big)\mathrm{d}u+\big(\partial_v\mathbf{n}+c\cdot\partial_v\mathbf{r}\big)\mathrm{d}v=\mathbf{0},$$

which means

$$\mathbf{n}+c\cdot\mathbf{r}\equiv\text{ constant vector field},$$

so there is a constant vector $\mathbf{r}_0$ such that

$$\mathbf{n}+c\cdot\mathbf{r}\triangleq c\cdot\mathbf{r}_0,$$

i.e.,

$$\big(\mathbf{r}-\mathbf{r}_0\big)^2=\frac{1}{c^2},$$

which is an equation for a sphere. ■